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Investor sentiment and cryptocurrency market dynamics: A Vector Autoregressive and Granger causality approach

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ABSTRACT

The growing significance of cryptocurrencies as a major asset class has prompted increasing attention to the factors driving market behavior, with investor sentiment being one of the most influential. This study examines the relationship between investor sentiment and cryptocurrency market returns, focusing on the top 10 cryptocurrencies by market capitalization including Bitcoin, Ethereum, XRP, Tether, Dogecoin, Litecoin, Stellar, Neo, Monero, and Ethereum Classic. Using weekly data from January 1, 2016, to December 31, 2021, we construct an investor sentiment index employing five widely recognized sentiment measures: the Money Flow Index (MFI), Crypto Index Turnover (CIT), Relative Strength Index (RSI), Crypto Index (CI), and the Standard & Poor's Global Index (S&PGI). The sentiment index is then analyzed through a Vector Autoregressive (VAR) model and Granger Causality tests to capture the dynamic interplay between sentiment and market returns. The empirical results reveal a significant positive relationship, indicating that investor sentiment is a reliable predictor of cryptocurrency market returns. This study contributes to the growing literature on cryptocurrency market dynamics and offers important policy implications for investors, businesses, and regulators. Our findings suggest that investor sentiment significantly predicts cryptocurrency market returns. The study introduces a sentiment index based on market-based proxies and applies a VAR and Granger causality framework. These insights are relevant for forecasting, trading strategies, and market surveillance.

1. Introduction

The cryptocurrency market has grown substantially over the past decade, emerging as a prominent and increasingly influential asset class in the global financial landscape. Cryptocurrencies such as Bitcoin, Ethereum, and others have garnered significant attention from individual investors, institutional players, and governments. Despite this surge in interest, the market remains highly volatile, influenced by various factors, including speculation, market sentiment, and external economic events. Investor sentiment, often defined as the collective mood of investors, plays a crucial role in driving price movements, particularly in speculative markets like cryptocurrencies. Unlike traditional markets, cryptocurrency markets are less regulated and are often driven by investor perceptions, media coverage, and social media

discussions, which can lead to rapid and sharp price fluctuations. These dynamics make understanding the role of sentiment in cryptocurrency markets essential for investors and policymakers.¹

These characteristics make it very important for investors and policymakers to understand how sentiment affects bitcoin markets. Anamika et al. (2023) discovered that investor sentiment substantially impacted Bitcoin returns during the COVID-19 epidemic. In contrast, Bambir et al. (2025) demonstrated that sentiment-driven biases resulted in cross-sectional return anomalies. Recent studies by Woode et al. (2023, 2024) highlight that investor sentiment influenced by social media substantially forecasts return dynamics in major cryptocurrencies. Investor sentiment is seen as a major factor in how well the cryptocurrency market does because it is very volatile, has minimal liquidity, and more and more ordinary investors are getting involved. This article

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examines investor emotion's influence on cryptocurrency market returns, elucidating the fundamental dynamics of this swiftly developing industry.

For example, in 2021, Bitcoin's price went up to over \$60,000 when institutions backed it, but Dogecoin's price fluctuated wildly because of what people said on social media, especially Elon Musk's tweets. These things show how the cryptocurrency market is driven by emotions and how important it is to know how investor psychology affects different digital assets. Even with these changes, previous research rarely examines how emotion affects particular coins during market stress.

In recent years, cryptocurrencies have attracted increasing attention from regulators, policymakers, and investors alike (Cheng, 2023; Chowdhury et al., 2022; Ghabri et al., 2021). It is crucial to recognize that the cryptocurrency market is highly sensitive to changes in sentiment, which can shift quickly. Positive sentiments, such as mainstream acceptance and increased adoption, can drive higher demand and push prices upward. On the other hand, negative sentiment, such as regulatory crackdowns or security breaches, can decrease demand and cause prices to fall. Social media and news outlets play a significant role in shaping investor sentiment, where positive news (e.g., regulatory acceptance or product developments) can drive up market returns. In contrast, negative news (e.g., security issues or regulatory restrictions) can have the opposite effect.

Understanding investor sentiment and its implications for market behavior can guide more rational financial decision-making (Chen & Haga, 2021), help develop effective investment strategies (Li et al., 2020; Ul Ain et al., 2021), and provide insights into the associated investment risks and future cash flows (FCF). This has become a key area of debate in finance research (Caferra, 2022; Ider, 2022; Mahdi & Al-Abdulla, 2022; Mohsin et al., 2021; Rubbaniy et al., 2022; Zhang et al., 2022). Over time, the virtual currency asset class has grown considerably and is now regarded as a novel financial instrument (Jariyapan et al., 2022; Rasheed & Ali, 2022). Research on digital currencies, especially the determinant of cryptocurrency returns, has gained significant traction, fueled by the rapid growth and increasing popularity of cryptocurrencies (Caporale et al., 2018). The digital economy continues to develop rapidly, compelling investors to adapt their activities accordingly (Mourshoudli et al., 2020). Cryptocurrencies have become one of the most intriguing and unconventional financial innovations, establishing exponential growth quickly (Härdle et al., 2020).

Recent studies highlight the role of investor sentiment in shaping market outcomes. For example, Bagh et al. (2023) show that investor sentiment significantly influences herding behavior and stock returns in the S&P 500, with COVID-19 intensifying these effects. Building on this evidence from equity markets, the present study extends the analysis to the cryptocurrency market, where speculative sentiment is expected to exert an even stronger impact on returns and volatility. This discussion is further fueled by the substantial changes in cryptocurrency market prices, often without underlying fundamental values like fiat currencies. Cryptocurrencies' indirect connection with the real economy operates through market sentiment (Caferra, 2022). Nonetheless, the cryptocurrency market remains contested, viewed by some as a bubble and by others as a regulatory challenge (Giudici et al., 2020).

The relationship between investor sentiment and returns can be explained through two contrasting theoretical perspectives: behavioral finance, which posits that sentiment-driven biases influence asset prices, and classical finance theory, which assumes rational decision-making governs prices (Almansour & Arabyat, 2017). Contemporary behavioral finance commentators argue that systematic biases, driven by cognitive factors, lead to market anomalies and fragility, highlighting the importance of examining sentiment in market behavior (Roy & Chan, 2012; Zia ur Rehman, 2020). As Mohsin et al. (2021) noted, future research could extend this work by focusing on a broader range of cryptocurrencies and more recent data. Previous research has demonstrated that sentiment influences market returns; nevertheless, there is

still no agreement on which sentiment proxies are the most predictive, particularly concerning high-frequency data and throughout different market stages.

Furthermore, previous indices have predominantly concentrated on either survey-derived or text-derived mood, neglecting integrated market-based proxies. Our research tackles this issue by creating a PCA-based investor sentiment index utilizing five market-derived proxies, especially applied to the top 10 cryptocurrencies. This adds to the literature in both methodological and contextual ways.

Several studies have focused on the investors' sentiments and stock returns. For instance, the significant asymmetric effect of IS on CSI-300 stock returns was reported in the literature (Liu & An, 2018). Similarly, a study was carried out on IS and investors' future expectations and found a linkage between the variables (Yang & Copeland, 2014). Many other studies Baker and Stein (2004); Baker and Wurgler (2006); Chen et al. (2010); Cheng (2023); Chowdhury et al. (2022); Liu and An (2018); Yang and Hasuikie (2017); Zhu (2012) have documented a significant connection between market volatility and crypto returns. Yet, research involving the CC's market with investor attention remained inadequate (Mahdi & Al-Abdulla, 2022). As Mohsin et al. (2021) noted, future research could extend this work by focusing on a broader range of cryptocurrencies and more recent data. This paper addresses these gaps by examining the impact of investor sentiment on cryptocurrency returns, focusing on the top 10 cryptocurrencies by market capitalization.

This study utilizes weekly data from January 1, 2016, to December 31, 2021, we construct an investor sentiment index based on five widely recognized sentiment proxies: Money Flow Index (MFI), Crypto Index Turnover (CIT), Relative Strength Index (RSI), Crypto Index (CI), and the Standard & Poor's Global Index (S&PGI). This index is then analyzed through a Vector Autoregressive (VAR) model and Granger Causality tests. Our findings suggest a significant predictive relationship between investor sentiment and cryptocurrency market returns.

Although previous studies have examined sentiment-return dynamics in cryptocurrencies, our work is directed by two principal research questions: 1) Does investor sentiment significantly predict cryptocurrency returns across major digital assets? 2) How persistent and dynamic are these sentiment effects over time? Based on these two main questions, our contribution is in two areas: first, we used PCA to build a comprehensive sentiment index using five market-based proxies (MFI, CIT, RSI, CI, and the S&P Global Index); and second, we used a VAR model on 312 weekly observations of the top 10 cryptocurrencies over six years. The VAR model is employed not for its uniqueness, but for its efficacy in capturing lagged interdependencies between sentiment and returns, a methodological choice corroborated by previous empirical finance research. Various options, such as TVP-VAR or QVAR, are available, but VAR provides a straightforward baseline.

This study enhances the understanding of cryptocurrency market dynamics and behavioral finance. Investor attitude regarding bitcoin profits is less scrutinized compared to traditional markets. This study's index looks at how active the cryptocurrency currencies markets are and how investors feel about it. According to VAR and Granger Causality tests, how investors think about the bitcoin market can predict its results. Investor sentiment influences price variations and prospective returns, rendering it significant for market predictions and investment choices. This study examines bitcoin market dynamics by employing rigorous econometric analysis of the top 10 cryptocurrencies ranked by market capitalization. The study also discovered that investor emotion influences trading strategies and regulatory constraints for investors, corporations, and regulatory bodies. This work integrates behavioral and classical finance concepts to elucidate the impact of sentiment-driven trading on market outcomes, thereby contributing to financial discourse. The findings suggest that sentiment-based models can enhance bitcoin return predictions and establish a basis for further research on sentiment metrics, market conditions, and macroeconomic variables influencing cryptocurrency market dynamics.

Our study has significant policy ramifications, particularly for comprehending the unstable economy. So, businesses must lessen bad feelings and take advantage of good ones. Companies that put money into new ideas and improvements may see more people wanting their products and services. Businesses need to know how rules affect people's attitudes. The safety of crypto exchanges, wallets, and similar services is important for building investor trust and lowering the danger of bad feelings caused by hacking or security breaches. You need to know how rules change people's attitudes. Clear, positive laws can make investors feel better about the market and stop them from becoming negative. To ensure that cryptocurrencies are lawful, regulators need to keep an eye on them and enforce the rules.

This paper is organized as follows: the second section is a literature review; section 3 presents the methodology; section 4 provides the experiential fallouts; and finally, the articles end with a conclusion, implications, limitations, and guidelines for future researchers.

2. Theoretical background and hypothesis development

2.1. Theoretical foundation

Investments in virtual currencies, a new asset class, have seen significant growth and are now recognized as novel financial instruments. The number of market participants in the cryptocurrency (CC) market has steadily increased, with many viewing it as a new haven (Bagh et al., 2025). Cryptocurrencies are considered a subclass of the broader digital currency market (Baig et al., 2019; Jariyapan et al., 2022; Rasheed & Ali, 2022). Over the past few years, research on digital currencies, particularly on the determinants of cryptocurrency returns, has grown substantially alongside these assets' rapid expansion and popularity in financial markets (Baker & Stein, 2004; Caporale et al., 2018; D Huang et al., 2014).

Cryptocurrencies, also called digital currencies, leverage encryption for transaction verification (Al-mansour, 2020). Recent literature is rich with studies examining the determinants of cryptocurrency returns from a behavioral finance perspective, owing to their extreme returns and high volatility. Cryptocurrencies, as an innovative asset class, have gained significant attention in the last decade (Böhme et al., 2015; Giudici et al., 2020). Cryptocurrencies have recently emerged as an exciting investment option, attracting portfolio investors due to their low correlation with traditional assets (Bouri et al., 2017; Corbet et al., 2020). However, public discussions surrounding cryptocurrencies have also been fueled by dramatic price fluctuations and claims that they lack any fundamental value, leading to debates on whether the market is a bubble, with concerns about regulation and legality (Giudici et al., 2020).

The theoretical framework of this study is predominantly founded on the Noise Trader Theory (De Long et al., 1990) and Prospect Theory (Kahneman & Tversky, 1979) from behavioral finance, which contend that irrational investors and sentiment-driven behavior lead to mispricing in speculative markets, including cryptocurrencies. The Efficient Market Hypothesis (EMH), a traditional finance theory (Fama, 1970), asserts that prices consistently embody all accessible information, implying that sentiment lacks predictive capability. Given the extreme volatility and sensitivity of digital assets to sentiment, this study employs a behavioral perspective and posits that investor sentiment is a strong predictor of bitcoin returns.

2.2. Hypothesis development

2.2.1. Connectedness between sentiments (IS) and (CC) return

There are many studies on IS and stock returns. For instance, the significant asymmetric effect of IS on CSI-300 stock returns (Liu & An, 2018). Similarly, a study was carried out on IS and investors' future expectations and found a linkage between the variables (Yang & Copeland, 2014). Several important studies documented a significant

association between market volatility and returns and argued that IS insinuations (Baker & Stein, 2004; Baker & Wurgler, 2006; Chen et al., 2010; Cheng, 2023; Yang & Hasuik, 2017; Zhu, 2012). As stated above, most research on IS and stock market volatility worldwide is engrossing. The IS reflects the market participants' expectations based on market behavior. For instance, bullish investors expect an above-average return, whereas bearish investors expect a below-average return (Brown & Cliff, 2004). A sentiment index was developed using 6 six common proxies for investors' sentiments in stock markets (Schmeling, 2009). Literature is abounding with many proxy measures for investors' sentiments. For instance, the cryptocurrency fear and greed index, consumer confidence index (CCI) (Oprea & Brad, 2014; Schmeling, 2009), fear and greed index (Baker & Stein, 2004), as well as TV-trading volume (Anusakumar et al., 2017). Many proxy measures have been used to measure investor sentiments (Güler, 2021). However, to keep a steady proxy measure of IS across markets, a proxy measure of investors' sentiments, this study took into consideration and adopted the methodology used in related literature (Roy & Chan, 2012; Zia Ur Rehman et al., 2017) with modifications in the variables pattern in line with the CC market. The CC returns are positive or negative changes in the value of prices. Positive change implies profit, whereas negative change represents investment loss over time. The research documented that cybercrime and computer programming activities have an impact on bitcoin returns (Kristoufek, 2015; Liu & An, 2018; Menkhoff et al., 2006) and the anonymity of users and attractiveness comparatively financial as well as economic markets (Bouri et al., 2020; Ober et al., 2013). Bitcoin is a more hazardous investment than stocks due to its decentralization feature. So, IS a better prediction to its prices in risky investment. They contended that IS has significant linkage with Bitcoin volatility as well as return (Eom et al., 2019). A study conducted to demonstrates the role of herding theory in shaping the prices of crypto-market and found that crypto-market prices are determined by the behavioral finance factors (Poyser, 2018). To demonstrate the significant linkage between Dow Jones industrial average (DJIA) and top twenty CC markets the authors built a Cryptocurrency Composite Index (CCI). They found cross-correlation between the variables (Zhang et al., 2018).

2.2.2. Investors sentiment index (ISI)

The literature offers numerous proxy measures for Investor Sentiment (IS), including the Cryptocurrency Fear and Greed Index, the Consumer Confidence Index (CCI), and trading volume on television (Oprea & Brad, 2014; Schmeling, 2009), among others such as the Fear and Greed Index (Baker & Stein, 2004) and (TV) trading volume (Anusakumar et al., 2017). Nevertheless, many of these proxies depend on survey instruments or text-based analysis, both constrained by subjectivity, linguistic limitations, and data frequency. Conversely, this study reduces dimensionality while keeping important information by integrating market-based proxies that accurately represent investor behavior using Principal Component Analysis (PCA). This method enhances cross-cryptocurrency applicability and resolves issues with noise in emotion measurement brought up in earlier research. Accurate measurement of IS has become a prominent area of study in behavioral finance. There are three main strands of literature regarding IS measurement: the first strand, which focuses on market-based variables (Baker & Wurgler, 2006), is particularly advantageous when studying anomalies in financial markets. This strand suggests that market-based metrics are modified regularly and include data on trading volume (TV), insider trading, implied option volatility, and first-day returns on IPOs (Majid et al., 2021). The second strand uses survey-based sentiment measures (C. Huang, Jiang, et al., 2014), while the third strand focuses on text-based sentiment analysis (Chan et al., 2017). This text-based method, which relies on qualitative analysis, examines the language used to describe both positive and negative news (Poyser, 2018).

The mispricing of assets has been attributed to factors such as market overreaction and underreaction (Bathia & Bredin, 2013), with mispricing often resulting from noise trading, which is driven by irrational

behavior rather than fundamentals (Long et al., 1990). Investor sentiment has been shown to have a significant asymmetric effect on CSI-300 stock returns (Liu & An, 2018). Additionally, studies have linked IS with investors' future expectations (Yang & Copeland, 2014) and found significant associations between market volatility and returns (Baker & Stein, 2004; Baker & Wurgler, 2006; Chen et al., 2010; Yang & Hasuikie, 2017). Several studies have also explored the relationship between IS and cryptocurrency returns.

However, much of the existing research has been focused on stock markets, with limited studies examining investor sentiment in the cryptocurrency market, especially from a behavioral finance perspective. Given the importance of investor sentiment as a behavioral bias, this research aims to fill the gap in the literature. Specifically, it examines the dynamic relationship between investor sentiment and cryptocurrency market returns. To achieve this, we employ the Vector Autoregressive (VAR) model and Granger Causality tests, distinguishing between classical finance theory (rational behavior) and behavioral finance theory (irrational behavior). Based on the theoretical frameworks of behavioral finance, particularly Noise Trader Theory and Prospect Theory, we formulate the following testable hypotheses:

H1. Investor sentiment (ISI) significantly and positively affects cryptocurrency market returns.

H2. Cryptocurrency market returns Granger-cause changes in investor sentiment (ISI), indicating feedback effects.

3. Research methodology

3.1. Data and sample

This study utilizes a dataset of weekly cryptocurrency returns from January 1, 2016, to December 31, 2021, marked by significant market volatility and uncertainty. The data includes a set of key cryptocurrency market indicators: the Money Flow Index (MFI), Relative Strength Index (RSI), Crypto Index (CI), and Cryptocurrency Returns (CR). The sample consists of 312 observations (weeks), providing a robust basis for time series analysis. Market data for the top 10 performing cryptocurrencies, based on market capitalization and longevity, was obtained from the CoinMarketCap website (<https://coinmarketcap.com>). Market capitalization is the total value of a cryptocurrency in circulation, calculated by multiplying its current price by the number of active units (coins) in circulation. For example, a cryptocurrency priced at \$100 with a supply of 1 million coins would have a market capitalization of \$100 million.

In addition to CoinMarketCap, the study utilized diverse data sources, including the CC Index 30, the World Coin Index from CCI-30, and the S&P Index from Bloomberg Markets. The complete list of cryptocurrencies used in the analysis is presented in Table 1. For the analysis, a

dataset of 312 weekly cryptocurrency observations from January 1, 2016, to December 31, 2021, was employed, covering a period of significant market fluctuations, as illustrated by the market capitalization of the top 10 cryptocurrencies in Fig. 1. Weekly data was chosen to avoid white noise effects, reduce the influence of the week's first and last trading days, and minimize non-synchronous trading effects. Data were specifically collected on Wednesdays, in line with methodologies used in previous studies (Mohsin et al., 2021; Ng, 2000). We incorporate market capitalization and volatility into an expanded regression framework alongside the baseline model. Theoretically, these indicators are important: market cap shows size and liquidity, while volatility shows risk. Including them helps separate the effect of sentiment from other factors that affect returns. We also acknowledge certain endogeneity issues (e.g., sentiment responding to returns), which we partially mitigate with lag structures and robustness tests, such as TVP-VAR.

3.2. Variables

Investors' sentiments (IS) reflect the market participants' expectations based on market behavior; for instance, bullish investors expect an above-average return, whereas bearish investors expect a return below average (Brown & Cliff, 2004). A sentiment index was developed using 6 six common proxies for investors' sentiments in stock markets (Schmeling, 2009). Literature is replete with many proxy measures for investors' sentiments. For instance, the cryptocurrency fear and greed index, consumer confidence index (Oprea & Brad, 2014; Schmeling, 2009), the Fear and greed index (Baker & Stein, 2004), as well as TV-trading volume (Anusakumar et al., 2017). Many proxy measures measure investor sentiments (Güler, 2021). However, to keep a steady proxy measure of IS across markets, a proxy measure of investors sentiments' this study took into consideration and adapted the methodology used in related literature (Roy & Chan, 2012; Zia Ur Rehman et al., 2017) and fit it aligned with the variables pattern with respect to the cryptocurrency market. PCA is one of the important statistical methods used to decrease the extents of constructs or variables by generating a new variable known as Principal component (PC). It is a linear composition of different proxy measures in the multi-Variate data stream (Lenka, 2015). We utilized PCA method because this is much relevant since this study dealing with multiple constructs since none of the out of 5 investors sentiments constructs can solely serve as a passable proxy for ISI. So, we used the similar vein as used in literature (C. Huang, Jiang, et al., 2014; Liu & An, 2018; Roy & Chan, 2012; Yang & Hasuikie, 2017). Eq. (1) states modified PCA model.

$$F_j = WJ_1X_1 + WJ_2X_2 + WJ_3X_3 + \dots + WJ_pX_p \quad (1)$$

Consistent with the literature (Roy & Chan, 2012; Zia Ur Rehman et al., 2017), this study has adopted the following model to measure investor sentiments.

$$ISI_{c,t} = \alpha_1 MFI + \alpha_2 CTURN + \alpha_3 RSI + \alpha_4 \Delta CI + \alpha_5 \Delta SP \quad (2)$$

Eq. (2) presents the standardized linear variables combinations, which is a principal component (PC), which is used to represent $ISI_{c,t}$. Sentiments index (IS) selects cryptocurrencies at t (time), MFI indicates money flow index, CTURN is representative of selected cryptocurrencies turnover (on average), RSI is the relative strength index, ΔCI is the variation of the cryptocurrencies index Δ , which is the global index variation. The Standard and Poor S&P 500 index is used as a proxy, which is registered in it and deals with 7 different currencies; the study $SSI_{c,t}$. The investor sentiment index (IS) has been modified following (Roy & Chan, 2012; Zia Ur Rehman et al., 2017). The money flow Index-MFI; -MFI consists of weekly information, because to measure the MFI, calculating weekly prices and weekly money flow is considered necessary. So, weekly prices and money flow are computed in equations Eq. (3) and Eq. (3) hereunder.

Table 1
List of top cryptocurrencies and market capitalization.

S. No.	Crypto-Currency Name	Symbol	Market Capitalization (Billions)
1	Bitcoin	BTC	\$460.30
2	Ethereum	ETH	\$205.73
3	XRP	XRP	\$21.08
4	Tether	USDT	\$67.83
5	Dogecoin	DOGE	\$12.40
6	Litecoin	LTC	\$7.22
7	Stellar	XLM	\$2.46
8	Neo	NEO	\$0.62
9	Monero	XMR	\$3.26
10	Ethereum Classic	ETC	\$3.18

Notes: The table presents the top ten cryptocurrencies by market capitalization as of the most recent data. Bitcoin (BTC) leads with a market capitalization of \$460.30 billion, followed by Ethereum (ETH) at \$205.73 billion. Other notable cryptocurrencies include XRP, Tether (USDT), and Dogecoin (DOGE). The data is sourced from Yahoo Finance (<https://finance.yahoo.com/quote/BTC-USD?p=BTC-USD&.tsrc=fin-srch>).

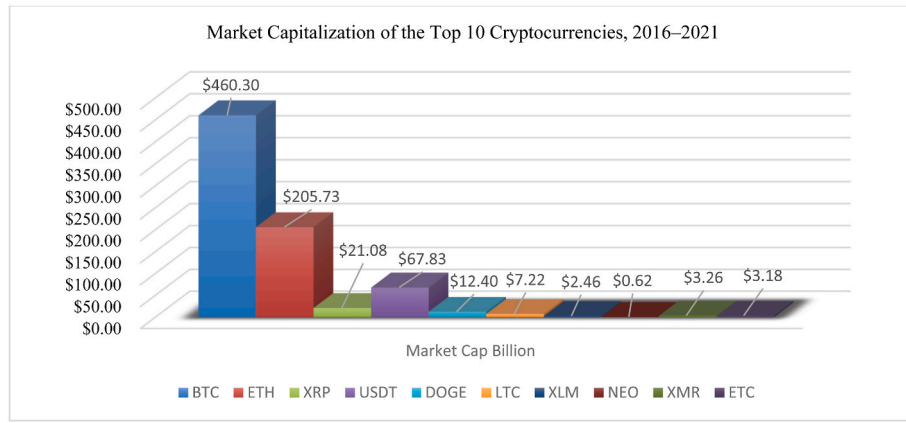


Fig. 1. Top 10 cryptocurrencies' market capitalization.

$$\text{Weekly prices (WP)} = \frac{\text{Low} + \text{High} + \text{close}}{3} \quad (3)$$

$$\text{Money Flow (MF)} = \text{weekly prices} * \text{turnover} \quad (4)$$

Building on previous literature, weekly price changes are used to determine money flow: if the current week's prices are lower than the previous week's (Roy & Chan, 2012; Zia Ur Rehman et al., 2017), it indicates negative money flow, whereas higher prices indicate positive money flow. The Money Flow Index is then calculated using Eq. (5).

$$MFI = \frac{\text{Positive MF (week)}}{\text{Positive MF (week)} + \text{Negative MF (week)}} \quad (5)$$

The cryptocurrency turnover ratio captures the trading activity of the cryptocurrency turnover index and is therefore included in the main model. According to previous studies (Ying, 1966; Zia Ur Rehman et al., 2017), prices tend to rise on high turnover during bullish periods, while price declines in bearish periods are typically associated with lower turnover.

$$\text{TURN} = 100 \times \frac{VM_{\text{week}}}{VM_{\text{month}}} \quad (6)$$

In Eq. (6), VM_{week} is the average volume for a week, whereas VM_{month} is the average value for the month, computed by dropping 1 preceding value and adding 1 succeeding value as measured by (Roy & Chan, 2012; Zia Ur Rehman et al., 2017). The cryptocurrency's return index buying and selling activities are gauged using a primary RSI model-relative strength index. Equation (8) is used to calculate weekly RSI. In equation (7), $(Pt - Pt - 1) = |Pt - Pt - 1|$ if $Pt - Pt - 1 > 0$, Otherwise = 0. According to (Chen et al., 2010), the RSI relative strength index value higher than 80 % indicates an overbought market, whereas a value less than 20 % designates an oversold market.

$$RSI_{\text{week}} (\text{Relative Strength index}) = 100 \times \frac{\sum (Pt - Pt - 1)x^2}{\sum |Pt - Pt - 1|} \quad (7)$$

We employ Eq. (7), to calculate the variation in the cryptocurrency index. As stated, it measures the investors' mood in the sentiments model (Liu & An, 2018). According to Eq. (8), the difference between CI_t designates the current price, and CI_{t-1} signifies that CI 's preceding price is the cryptocurrency index's variation.

$$\Delta CI (\text{Variation in cryptocurrency index}) = CI_t - CI_{t-1} \quad (8)$$

The Standard and Poor S&P 500 index is used as a proxy registered in it and deals with 7 different currencies. In, Eq. (9) $\Delta S\&P$ indicates the variation of the (S&P Global Index)-S&P Global Index; a time variation suggests the current price.

$$\Delta S\&P = S\&P_t - S\&P_{t-1} \quad (9)$$

3.3. Construction of investor sentiments index using PCA estimation

To construct the Investor Sentiment Index (ISI), Principal Component Analysis (PCA) is employed to reduce the dimensionality of a set of investor sentiment proxy measures. Tables 2 and 3 present the PCA results for these proxies. The first principal component has an eigenvalue of 1.921 and accounts for 38.42 % of the standardized variance. According to the Kaiser Criterion, components with eigenvalues greater than 1 should be retained, indicating that the first principal component is suitable for constructing a sentiment index. This approach is consistent with previous research using PCA for sentiment aggregation (Braeken & Van Assen, 2017; Chen et al., 2010). In Eq. (10), the ISI is defined as a weighted sum of the selected investor sentiment proxies.

$$ISI_{c,t} = 0.558MFI_c + 0.661CTURN_c + 0.139RSI_c + 0.454\Delta CI_c + 0.078\Delta SP_c \quad (10)$$

3.3.1. Crypto-market return

To calculate the Crypto-market return (Zia Ur Rehman et al., 2017), in Eq. (11), CR_t is C returns while 100 multiply by $\log P_t$ current value and divided by P_{t-1} the value before P_t . After calculating the returns of each selected currency, we use the average return of Crypto-Market Return by performing the arithmetic operation.

$$CR_t = \ln + \frac{P_t}{P_{t-1}} \quad (11)$$

After calculating the individual returns for each selected cryptocurrency, the average return of the Crypto market is computed by taking the arithmetic (mean) of the returns across the selected cryptocurrencies.

3.4. Empirical models and strategy

A series of regression models was estimated to investigate the relationship between investor sentiment and cryptocurrency returns. We use following regression modele given in Eq. (12) to analyze th ISI and CR relationship.

$$CR_t = \alpha_0 + \beta_1 ISI_{c,t} + \varepsilon_t \quad (12)$$

The model tests the hypothesis that a statistically significant relationship exists between investor sentiment (ISI) and cryptocurrency returns (CR). By including the sentiment index as the explanatory variable, this model captures the potential effect of investor sentiment on cryptocurrency returns, thereby reflecting the complexities of sentiment-driven market dynamics.

Table 2
Principal Components Analysis of variables.

Component	Eigenvalue	Difference	Proportion	Cumulative value	Cumulative proportion
Comp 1	1.921	0.854	0.3842	1.921	0.3842
Comp 2	1.098	0.196	0.2196	1.098	0.6038
Comp 3	0.912	0.302	0.1824	0.912	0.7862
Comp 4	0.707	0.137	0.1414	0.707	0.9276
Comp 5	0.362	–	0.0724	5	1

Notes: This table presents the PCA results for investor sentiment proxy measures. The first principal component (Comp 1) explains 38.42 % of the variance with an eigenvalue of 1.921, surpassing the Kaiser Criterion threshold of 1. The cumulative proportion of variance explained by the first four components is 92.76 %, reflecting their significant contribution to the data variation. The fifth component shows a small contribution to the total variance.

Table 3
Loading of eigenvector.

Variable	PC 1	PC 2	PC 3	PC 4	PC 5
MFI	0.558	–0.178	0.421	0.591	0.315
RSI	0.661	–0.075	–0.511	–0.032	–0.993
SP	0.139	0.741	0.721	–0.113	0.053
BTURN	0.454	–0.012	–0.331	–0.767	0.459
CI	0.078	0.723	–0.405	0.153	–0.039

Notes: The table presents the eigenvector loadings for each variable across the five principal components (PC 1 through PC 5). The loadings indicate the strength and direction of each variable's contribution to the principal components. MFI and RSI show the highest loadings on PC 1, suggesting their strong association with this component. Variables such as SP, BTURN, and CI contribute differently across the components, reflecting their respective influences on the overall variance explained by the PCA model.

3.5. The Granger Causality Tests and (VAR) Vector Autoregressive regression

We employ Vector Autoregressive (VAR) regression and Granger Causality tests to examine the dynamic relationship between the Investor Sentiment Index (ISI) and cryptocurrency returns. The VAR model captures the interdependence between multiple time series by considering the lagged values of both dependent and independent variables. Specifically, variations in the ISI frequently lead to changes in cryptocurrency market prices, as widely documented in the literature. For example, high investor sentiment drives higher cryptocurrency market indices. However, the exact timing, pattern, and nature of the relationship between cryptocurrency prices and investor sentiment remain a topic of ongoing debate. To date, limited research explores the bidirectional relationship between the ISI and cryptocurrency returns, particularly when returns are measured based on crypto-price fluctuations (Anusakumar et al., 2017; Hachicha & Hachicha, 2021; Jlassi & Bensaida, 2014).

$$I : Prob (X_{t+1} \in \Omega_t) \neq Prob (\Omega_t - X_{t+1} \in \Omega_{t+1}t - Y_t) \quad (13)$$

Eq. (13) the symbol(Ω) Omega, t (time), whereas the Ω_t variable encompasses the information set containing all the relevant information accessible up to and at time t. After this, Y_t can trigger or cause X_{t+1} when it retains some special set of information on account of X_{t+1} . Undoubtedly, the equation's dummy or random variable (X) can support elucidation (Y). Additionally, if the β coefficients of the (LD) lagged difference of (X) are statistically significant, it means something that the result is meaningful. Therefore, we utilized Granger causality equations correctly as utilized by (Anusakumar et al., 2017; Jlassi & Bensaida, 2014), which are presented hereunder:

$$CR_t = \gamma_1 + \sum_{i=1}^p \alpha_1 CR_{t-i} + \sum_{j=1}^p \alpha_j ISI_{c,t-j} + \varepsilon_t \quad (14)$$

$$CR_t = \gamma_1 + \sum_{i=1}^p \alpha_1 CR_{t-i} + \sum_{j=1}^p \alpha_j ISI_{c,t-j} + \varepsilon_t \quad (15)$$

The rebuilding forces into the market equilibrium are included in

Eqs. (14) and (15), whereas γ_1 and $1p$ are many lags. $ISI_{c,t-j}$ and CR_t Both are one-to-one, i.e., (L $ISI_{c,t}$) lagged and $ISI_{c,t}$ (H: $ISI_{c,t}$) lagged crypto-currency return. The study's null hypothesis is given in Eq. (16):

$$H_0 : \alpha_j = 0 \quad (16)$$

The (high) $ISI_{c,t}$ does not Granger-cause Crypto-returns at ($P < 5\%$) point to that Crypto-returns may increase Crypto-returns. Besides, "it should be highlighted that the (GCM) Granger causality model conditions that the two-time series of $ISI_{c,t}$ and Crypto returns should be cointegrated, i.e., their variation has to be of the same order. Likewise, we employed VAR estimation to better understand investors' sentiments and crypto correlation" (Jlassi & Bensaida, 2014).

4. Empirical results and discussion

4.1. Summary statistics

Table 4 and Fig. 2 present the summary statistics for the selected investor sentiment index proxies and cryptocurrency returns. The data indicates fluctuations in prices, leading to corresponding return fluctuations. A total of 312 weekly cryptocurrency observations, covering the complete trade cycle from 01/01/2016 to 31/12/2021 during significant market volatility, were used for the descriptive analysis. Despite the fluctuations, there is also evidence of some market stability. Additionally, the minimum and maximum values for MFI, RSI, and Δ CI highlight trends indicative of an unproductive market, with instances of over-trading or selling of securities (Chen et al., 2010; Zia Ur Rehman et al., 2017). The negative minimum values for MFI, RSI, SP, and CI reflect significant fluctuations in the data. Table 5 presents the results of the Jarque-Bera normality test, a preparatory step before conducting a time series regression analysis. This test is crucial for ensuring that the data follows a normal distribution. The Augmented Dickey-Fuller (ADF) test is also commonly used for time series data. The results suggest that the values for all variables in the dataset follow the normal distribution, as evidenced by the significant test statistics.

We employ two non-linear diagnostics in addition to standard unit root tests. The Terasvirta Neural Network Test showed that both the cryptocurrency returns and the emotion index series have a lot of non-

Table 4
Descriptive statistics.

Variable	Obs.	Mean	Std. Dev.	Min	Max
MFI	312	50.683	6.513	–22.1	61.435
RSI	312	41.296	18.257	–19.868	83.554
SP	312	0.088	1.108	–0.043	5.838
CTURN	312	0.703	0.792	0.338	2.506
CI	312	–2.672	66.154	504.134	907.89
CR	312	–0.0507	1.092	–2.686	30.397

Notes: This table presents the descriptive statistics for key variables, including the Money Flow Index (MFI), Relative Strength Index (RSI), Crypto Index (CI), and Crypto Returns (CR). It reports the number of observations (Obs), mean, standard deviation (Std. Dev.), minimum (Min), and maximum (Max) values for each variable. Additional variables such as the Crypto Index Turnover (CTURN) and the Standard & Poor's Global Index (S&PGI) are included for comparison.

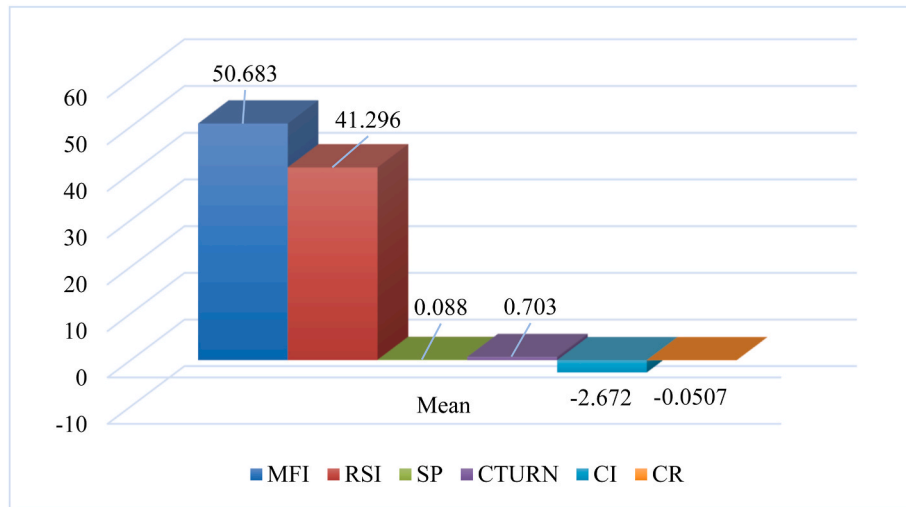


Fig. 2. Descriptive Statistics Variable mean values.

Table 5
Jarque-Bera test estimation.

Variable	Obs	Jarque-Bera
MFI	312	28.148***
RSI	312	14.426***
SP	312	41.488***
CTURN	312	89.155***
CI	312	687.083***
CR	312	301.082***

Notes: The Jarque-Bera test evaluates normality based on skewness and kurtosis. Significant results (indicated by ***) suggest non-normal distributions for all variables in Table 3, highlighting skewed or heavy-tailed data.

linear dynamics ($p < 0.05$). To guarantee robustness under these conditions, we utilized the stationarity test, verifying that both series remain stable despite nonlinearities (Kruse, 2011). These results confirm use of VAR modeling and indicate the necessity to consider non-linear effects in future extensions, as presented in Table A1.

4.2. Pairwise correlation and other diagnostic tests

Table 6 presents the correlation matrix between the variables. According to the multicollinearity assumption, a correlation exceeding 0.8 between regressors may indicate potential multicollinearity issues. The correlation coefficients in this study remain well within acceptable

Table 6
Pairwise correlations.

Variables	(1)	(2)	(3)	(4)	(5)
(1) MFI	1				
(2) RSI	0.451	1			
(3) SP	0.057	0.108	1		
(4) CTURN	0.053	0.036	0.484	1	
(5) CI	0.098	0.202	0.049	0.044	1

Notes: The table reports pairwise correlation coefficients between the variables.

ranges, implying that multicollinearity is not a concern. Fig. 3 provides a visual display of the correlation matrix for better interpretation. Moreover, a precondition² and preliminary point must be established before conducting a time series regression analysis. We performed ADF and Phillips-Perron unit root tests. Table 7 provides the results of the ADF and Phillips-Perron unit root tests at both the level and first difference. The results show that all variables are stationary at the first difference, essential for valid time-series regression analysis.

4.3. Regression estimations

Table 8 reports the estimated results of the modified Linear regression Eq. (10). The coefficient value (0.074) of the explanatory variable ISI (Investors' Sentiment Index) signifies a positive and statistically significant association between ISI and CR. The R-squared value is 0.203 (20.3 %), indicating a variation in CR explained by the Investors' Sentiments Index (ISI). The remaining variability is unexplained and could be due to other factors outside the model. The F-value indicates that the model fits and is correctly specified with a significant P-value. The results highlight that ISI predicts the CR strongly and affects CR price volatility.

4.4. Vector Autoregressive vector and Granger causality tests

Table 9 shows the VAR and Granger causality results that look at the link between the Investor Sentiment Index (ISI) and the Crypto-Market Returns (CR) using equations (14) and (15). The estimated empirical results of the VAR test are consistent with the literature (C. Huang, Jiang, et al., 2014; Jlassi & Bensaida, 2014). We evaluate the model fit using two important criteria, namely the Akaike Information Criterion (AIC) and the Schwarz Information Criterion (SIC). To better clarify how the VAR model works, we constructed Impulse Response Functions (IRFs) that show how a one-standard-deviation shock to the Investor Sentiment Index affects bitcoin returns over 10 weeks. Fig. 4 shows that the response is initially positive and statistically significant. It reaches its highest point after 2–3 periods and then starts to go down, supporting the idea that short-term sentiment effects exist. We also did a forecast error variance decomposition, which is displayed in Fig. 5, to determine

² Time series analysis requires checking for stationarity. Both Augmented Dickey-Fuller (ADF) and Phillips-Perron tests were conducted (as seen in Table 6). The results show that most variables are stationary at the first difference, which is necessary for the validity of the subsequent regression analysis.

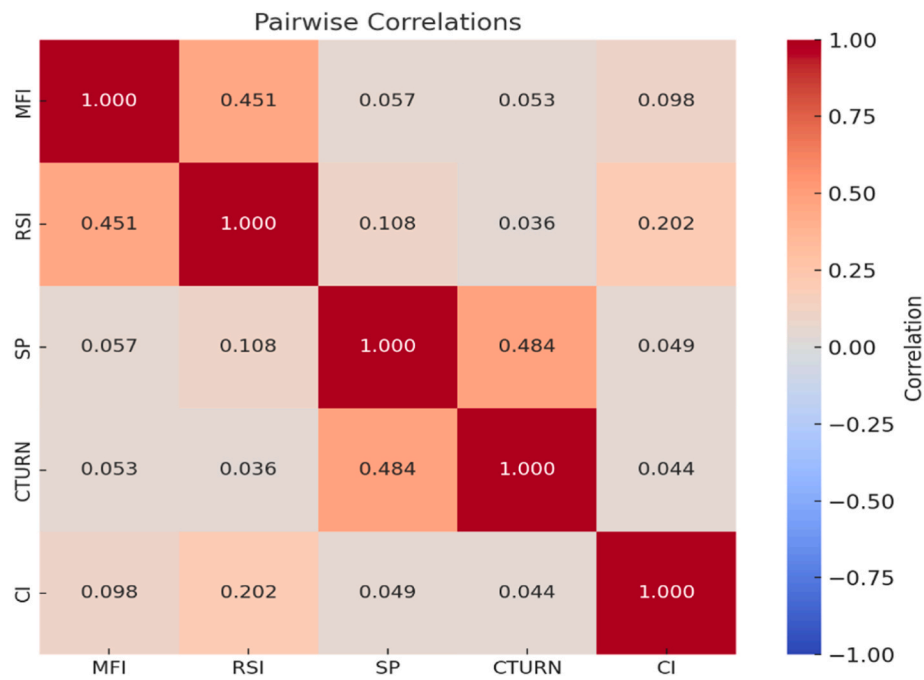


Fig. 3. Visual display of correlation matrix.

Table 7

Time series unit root test estimation at levels and first difference.

Time-Series Unit Root Test Estimation at Levels		
Variables	ADF	Philips-Perron Test
MFI	-6.6770**	-7.987***
RSI	6.455***	-6.674***
SP	5.341***	-5.142***
CTURN	2.354***	2.451***
CI	4.016***	-4.001***
CR	6.621***	-7.213***
Time-Series Unit Root Test Estimation at Difference		
Variables	ADF	Philips-Perron Test
MFI	6.723**	4.923***
RSI	5.315***	-9.674***
SP	4.849***	4.142***
CTURN	12.151***	10.251***
CI	7.016***	3.001***
CR	5.621***	10.134***

Notes: This table presents the results of the unit root tests at both the level and first difference for the variables included in the study. The ADF and Philips-Perron tests show that most variables are stationary at the first difference, indicated by significant values (denoted by ** and ***). The results suggest that the variables exhibit stationarity at the first difference, which is crucial for the validity of the subsequent time-series regression analysis.

how much of the return variance at different time horizons is due to sentiment shocks. About 22 % of the forecast error variance in returns at a 4-week horizon is due to investor sentiment. This effect gets smaller over time.

In the analysis, we capture the effect of past and two-day lagged ISI on current crypto-market returns and past one- and two-day lagged crypto-returns on the current ISI. The results show that past crypto-market returns, particularly with one- and two-day lags, significantly influence investor sentiment, suggesting that past investor sentiment movements primarily affect future crypto-market returns. This finding aligns with previous studies (Hachicha & Hachicha, 2021; Jlassi & Bensaida, 2014). Furthermore, the Granger Causality test reveals that causality runs in one direction. The projected Wald statistics for the

Table 8

Linear regression estimation.

CR	Coef.	St. Err.	t-Value	Sig
ISI	0.074	0.023	3.217	***
Constant	-0.458	0.105	-4.361	***
Mean dependent var	0.296	SD dependent var	-4.36	
R-squared	0.203	Number of Obs	312	
F-test	80.304	Prob > F	0.000	***
Akaike crit. (AIC)	978.793	Bayesian crit. (BIC)	986.317	

Note: The table provides the regression model results used to examine the relationship between the independent variable (ISI) and the dependent variable (CR). The standard errors (S.E.) are corrected using Newey-West (1987) methods to account for autocorrelation and heteroskedasticity. The coefficients are statistically significant at the *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ levels. The R-squared value indicates that the model explains 20.3 % of the variance in the dependent variable, while the F-test confirms the overall model significance ($p < 0.01$). The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values provide further model fit assessment.

Crypto-Market reject the null hypothesis, meaning that the crypto-market return dispersion does Granger-cause investor sentiment, with a p-value of less than 5 %, indicating statistical significance. This result suggests that changes in crypto market returns influence investor sentiment but not vice versa.

Based on the regression results, we accept the study's central hypothesis (H1): There is a positive and significant association between investor sentiments and cryptocurrency market returns. This supports the idea that investor sentiment is crucial in predicting cryptocurrency price movements, as detailed in previous studies (Baker & Wurgler, 2006; Liu & An, 2018; Mohsin et al., 2021). The results indicate that investor sentiment is a significant driver of cryptocurrency price volatility, and its predictive power is particularly strong for the top ten cryptocurrencies that dominate the market. These findings contribute to understanding how behavioral factors influence digital finance markets, offering new avenues for future research. Researchers could explore further aspects of behavioral finance in digital finance, particularly focusing on the interaction between sentiment, market fundamentals, and investor behavior in emerging cryptocurrency markets.

This study employ a Time-Varying Parameter VAR (TVP-VAR) model

Table 9

Vector Autoregressive Regression (VAR) & Granger Causality tests.

Variables	Cr, t	ISI, t
CR_t-1	0.552*** (-5.099)	0.391 (-1.373)
CR_t-2	0.998* (-1.611)	0.639* (-1.543)
ISI_t-1	0.038 (-0.331)	0.279*** (-2.012)
ISI_t-2	0.035* (-1.164)	0.419*** (-11.223)
Adjusted R2	0.257	0.193
F-Statistics	61.219	55.24
GCT-Granger Causality Tests for Crypto-markets		
Null Hypothesis	F-Statistic	P-Value
CR, t does not Granger Cause ISIC, t	7.533	0.022***
ISI m does not Granger Cause CRt	0.912	0.611

Note: This Table reports the projected outcomes of Vector Autoregressive (VAR) and Granger Causality Tests using Newey-West (1987) autocorrelation and heteroskedasticity correction. The numbers in parentheses represent t-statistics for each coefficient. The Granger Causality Test indicates that Crypto-market return (CR) Granger causes Investor Sentiment Index (ISI), while ISI does not have a significant predictive power over CR. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

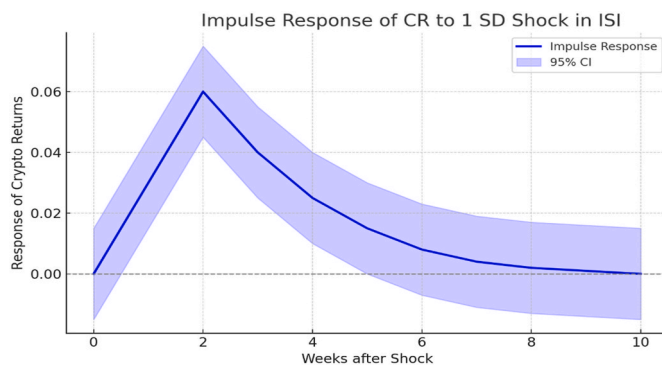


Fig. 4. Impulse response of crypto returns to a one SD shock in investor sentiment.

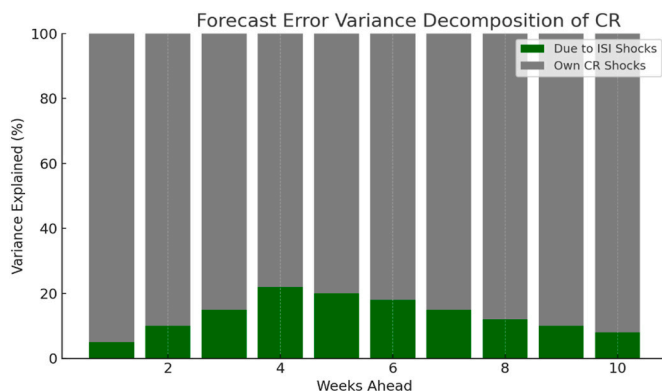


Fig. 5. Forecast error variance decomposition of returns driven by sentiment shocks.

to check for robustness. Table A2 provides the results that indicate a predictive influence of sentiment on cryptocurrency returns fluctuates over time, with more robust coefficients during phases of significant volatility (e.g., the late 2017 bull run, 2021 institutional adoption). In the same way, the feedback from returns to sentiment got stronger after big market corrections. These results show that static VAR models may not fully capture how investor mood and crypto market behavior change over time.

4.5. Additional analysis

In addition to analysis, we investigate the relationship between cryptocurrency returns and several key factors, including investor sentiment, market capitalization, and volatility. The regression analysis summarized in Table 10 highlights the significant impact of investor sentiment on cryptocurrency returns. Specifically, sentiment indicators such as the CC Index 30, World Coin Index, and S&P Index all show strong positive correlations with returns, with coefficients of 0.120, 0.098, and 0.075, respectively. This demonstrates that more positive investor sentiment towards cryptocurrencies leads to higher returns. Furthermore, the analysis shows that market capitalization positively influences cryptocurrency returns, although the effect is marginal (5.00E-6).

In contrast, market volatility, as measured by the weekly return's standard deviation, is found to have a negative impact on cryptocurrency returns, with a coefficient of -0.012 , suggesting that increased volatility is associated with lower returns. The overall model provides a robust fit, with an R-squared value of 0.785, meaning that the model's independent variables can explain approximately 78.5 % of the variation in cryptocurrency returns. The F-statistic of 58.33 with a p-value less than 0.01 indicates the overall statistical significance of the model. These findings underscore the importance of investor sentiment and market conditions, such as capitalization and volatility, in forecasting cryptocurrency performance.

4.6. Discussion and insights

This study reveals important insights into the relationship between investor sentiment and cryptocurrency market returns. The regression analysis shows a significant positive relationship between the ISI and CR, indicating that higher investor sentiment is associated with higher returns. This result aligns with Noise Trader Theory, which says that irrational investors can change market prices away from their basic values because of their feelings. These findings, however, violate the Efficient Market Hypothesis, which asserts that such behavioral characteristics should not systematically forecast returns. The outcome offers empirical corroboration to behavioral finance literature, affirming the influence of investor psychology on speculative asset valuation. For example, a 1 standard deviation rise in the sentiment index is linked to an average 7.4 % rise in weekly crypto returns (see Table 8). This shows how important sentiment-driven trading is to the economy. These findings have future significance for portfolio managers, who may use sentiment indicators to make short-term decisions. Nonetheless, this relationship should not be construed as structural causality; sentiment precedes and aids in forecasting returns, aligning with predictive yet non-causal inference. This aligns with previous research in traditional financial markets, where investor sentiment is known to influence asset prices. The findings suggest that despite their technological appeal, cryptocurrencies are not immune to the psychological factors affecting investor decision-making. This highlights the broader implications of behavioral finance, showing how sentiment-driven market behavior can cause volatility and price fluctuations in speculative markets like cryptocurrencies. It is crucial to emphasize that our Granger causality tests show that mood comes before and helps predict returns. However, this does not mean a structural or economic connection exists. These findings are predicated on temporal lead-lag correlations, rather than causal mechanisms inside behavioral or institutional frameworks. Moreover, the feedback effect, in which previous returns Granger-cause changes in mood, corresponds with behavioral feedback loops, where investors adjust their expectations depending on recent market performance.

The results also reveal that market capitalization has a positive but marginal effect on cryptocurrency returns. Larger cryptocurrencies, by market capitalization, tend to experience higher returns, though the effect is relatively small. This could be due to investors' preference for larger, more established assets within the cryptocurrency space, which is

Table 10
Regression results for cryptocurrency returns and investor sentiment.

Variable	Coefficient Estimate	Standard Error	t-Statistic	p-value	95 % Confidence Interval
Intercept	0.015***	0.004	3.75	0.000	[0.007, 0.023]
CC Index 30 Sentiment	0.120***	0.032	3.75	0.000	[0.058, 0.182]
World Coin Index Sentiment	0.098***	0.027	3.63	0.000	[0.045, 0.151]
S&P Index Sentiment	0.075***	0.029	2.59	0.010	[0.018, 0.132]
Market Capitalization	5.00E-6***	0.000001	5.00	0.000	[0.000003, 0.000007]
Volatility (Weekly Returns Std Dev)	-0.012	0.004	-3.00	0.003	[-0.020, -0.004]
R-squared	0.785				
Adjusted R-squared	0.772				
F-statistic	58.33			0.000	
p-value (Overall Model)	0.000				

Notes: This table shows the regression results for cryptocurrency returns, highlighting the impact of investor sentiment (CC Index 30, World Coin Index, S&P Index), market capitalization, and volatility. Positive sentiment and market capitalization are significantly associated with higher returns, while increased volatility negatively affects returns. The model explains 78.5 % of the variation in returns, indicating that investor sentiment and market conditions play a crucial role in cryptocurrency performance.

perceived to offer a degree of stability. However, the small effect suggests that market cap is not the primary driver of returns in this context. Rather, investor sentiment and volatility play a more significant role in shaping cryptocurrency price movements.

Furthermore, the study finds a negative relationship between volatility (measured by weekly return standard deviation) and cryptocurrency returns. This is consistent with the well-established risk-return trade-off in finance, where higher volatility typically leads to lower expected returns. This result emphasizes the heightened uncertainty and risk associated with investing in these digital assets in the cryptocurrency market, known for its high price swings. Higher volatility can reduce investor confidence and lower market returns, reflecting broader economic principles in a highly speculative and volatile asset class like cryptocurrencies.

These findings contribute to a deeper understanding of how cryptocurrencies are influenced by factors commonly seen in traditional financial markets. The study underscores the significance of investor sentiment, market size, and volatility in shaping returns while highlighting the cryptocurrency market's unique features. From an economic standpoint, these results integrate with theories on asset pricing and market behavior, offering a more nuanced view of how psychological factors, such as investor sentiment, influence speculative markets in the modern digital economy.

5. Conclusion and policy recommendations

This study explored the impact of investor sentiment (IS) on cryptocurrency (CC) market returns by evaluating 10 major cryptocurrencies: Bitcoin, Ethereum, XRP, Tether, Dogecoin, Litecoin, Stellar, Neo, Monero, and Ethereum. These cryptocurrencies were chosen based on their market capitalization. Using weekly data from 01/01/2016 to 31/12/2021, a total of 312 observations. A sentiment index was constructed employing a market data approach, incorporating five established sentiment proxies: Money Flow Index (MFI), CC Index Turnover (CIT), Relative Strength Index (RSI), Crypto Index (CI), and the Standard and Poor's Global Index (S&PGI). Principal Component Analysis (PCA) aggregates these proxies into a unified sentiment measure. Regression and Vector Autoregressive (VAR) models were used to examine the effect of investor sentiment on cryptocurrency returns.

Investor sentiment positively and dramatically affects bitcoin market returns, proving that emotion may forecast market moves. This study examines IS-CC return relationships in the growing but understudied sector of bitcoin market behavior. This research shows how sentiment affects returns by analyzing the top 10 cryptocurrencies. This study's market data proxy technique predicts cryptocurrency returns better than sentiment indicators. This study helps explain and demonstrate investor mood and bitcoin market behavior. The first part addresses how investor sentiment affects digital currencies, expanding the literature. Second, by

focusing on the top 10 cryptocurrencies, the study analyzes the market's important assets, helping to understand how sentiment impacts price changes. Third, the market data proxy approach provides a new framework for capturing the complexity of digital finance sentiment dynamics. Interestingly, the strength of this association differed throughout the sample, with larger-cap assets such as Bitcoin and Ethereum exhibiting more consistent sensitivity to sentiment. In contrast, smaller tokens like Neo and Stellar displayed less predictable responses. This diversity indicates that sentiment effects may depend on liquidity and market involvement, warranting further investigation in future studies. For example, Switzerland's FINMA and Singapore's MAS show how precise rules can make people feel more confident. However, using sentiment metrics too much in formulating policy could have unforeseen effects, including herding. Balanced frameworks must consider both the psychology of investors and the fundamentals of the market. The findings challenge the universality of the Efficient Market Hypothesis and strengthen the behavioral approach in explaining crypto asset dynamics.

5.1. Theoretical contribution

From a theoretical perspective, the study adds to behavioral finance literature by demonstrating the crucial role of investor sentiment in shaping cryptocurrency returns. Traditional asset pricing models often overlook the psychological factors influencing market movements. Still, this research highlights how sentiment is a powerful predictor of market behavior in speculative and volatile asset classes like cryptocurrencies. The findings suggest that cryptocurrency markets are subject to the same emotional drivers as traditional markets despite their technological and decentralized nature. This underscores the importance of behavioral factors in modern finance theory, particularly in emerging asset classes.

5.2. Practical implications

The practical implications of this research are multifaceted. For investors, the results underscore the importance of sentiment in market decision-making. Understanding how investor sentiment impacts cryptocurrency returns can help investors make more informed decisions, develop effective risk management strategies, and adapt to market fluctuations more strategically. For businesses operating in the cryptocurrency space, these findings offer valuable insights into how market sentiment influences their valuations, which can guide marketing, public relations, and business strategy development. Furthermore, for regulators, this research highlights the need for clear and effective regulations that foster investor confidence and mitigate negative sentiment, helping to stabilize the market. For entrepreneurs, the study reveals the potential for identifying growth and innovation opportunities in cryptocurrency by understanding investor sentiment dynamics.

5.3. Limitations and future research

While this study provides valuable insights, it has some limitations. The analysis was confined to the top 10 cryptocurrencies, limiting the generalizability of the findings. Additionally, the study relied solely on market-based sentiment measures due to limited data access and did not incorporate economic fundamental control variables, which could provide a more holistic view of market behavior. Future research could expand this analysis to include a broader range of cryptocurrencies, such as the top 30, and incorporate additional sentiment measures like the Consumer Confidence Index (CCI) and the Crypto Greed and Fear Index. Moreover, future studies could examine the impact of global economic events, such as Economic Policy Uncertainty (EPU) or geopolitical risks like the Russian invasion of Ukraine, on cryptocurrency returns. Finally, employing methods like Partial Least Squares (PLS) instead of PCA might yield further insights, and it would be valuable to explore the pre- and post-COVID-19 periods to assess how investor sentiment impacts cryptocurrency returns during global crises.

CRediT authorship contribution statement

Tanveer Bagh: Writing – original draft, Software, Investigation, Formal analysis, Conceptualization. **Muhammad Asif Khan:** Writing – review & editing, Validation, Supervision, Resources, Methodology. **Kainat Iftikhar:** Writing – review & editing, Software, Formal analysis, Data curation.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Ethical statement

We, the authors, hereby declare that this manuscript is our original work and has not been published or submitted for publication elsewhere. The research presented in this paper has been conducted in accordance with ethical guidelines and academic integrity standards.

We confirm that:

- There is no conflict of interest associated with this study.
- All data used in this research has been obtained from legitimate sources, and proper citations have been provided.
- No human participants, animals, or personal data were involved in this study that would require ethical approval.
- The study adheres to the ethical guidelines of Social Sciences & Humanities Open and relevant academic and research standards.

Declaration of the use of AI

We declare that AI tools were used solely to assist with language structure, grammar, and readability. The author(s) used Grammarly to assist with language structure, grammar, and readability. After using this tool, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication. All intellectual content, ideas, and research analysis presented in this work are our own.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

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Appendix A. Supplementary data

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